

11

Magnetic helicity interpreted and Woltjer–Taylor relaxation

11.1 Introduction

The previous chapter introduced the concept of magnetic helicity via the energy principle and showed that total helicity $K = \int d^3r \mathbf{A} \cdot \mathbf{B}$ is a conserved quantity in an ideal plasma. This chapter shows that helicity can be interpreted in a topological sense as a count of the linkages of magnetic flux tubes with each other. Furthermore, it will be shown that when the plasma is not ideal so energy is not conserved, helicity conservation remains a rather good approximation.

The greater robustness of magnetic helicity compared to magnetic energy in the presence of dissipation leads to the Woltjer–Taylor relaxation theory, which shows that a dissipative plasma will spontaneously relax from an arbitrary initial state to a specific final state. Relaxation theory has two remarkable features, namely (i) it sidesteps describing the actual MHD dynamics and simply predicts the end state after all dynamics is over, and (ii) it thrives on complexity so the more complicated the dynamics, the more applicable is the theory. The second feature results because increased complication simply provides more channels whereby the plasma can relax to the specific final state. Relaxation theory has been very successful at predicting the approximate behavior of many laboratory, space, and astrophysical plasmas.

This chapter concludes by showing that magnetic helicity can be manifested in different forms. In particular, the kink instability will be shown to be a mechanism that converts helicity from one of these forms (twist) to another (writhe).

11.2 Topological interpretation of magnetic helicity

11.2.1 Linkage helicity

Consider the two thin linked untwisted flux tubes sketched in Fig. 11.1. The respective fluxes of these two tubes are Φ_1 and Φ_2 , the flux tube axes follow the

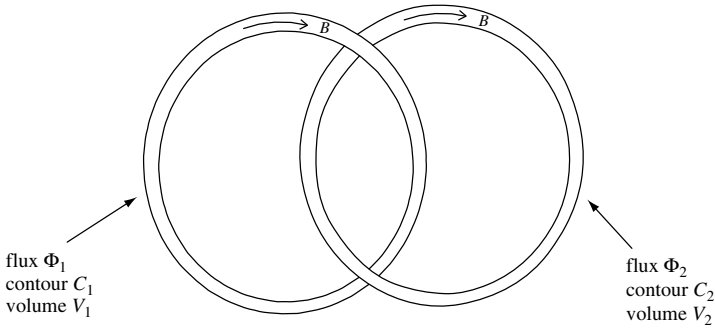


Fig. 11.1 Two linked thin untwisted flux tubes. Magnetic field is zero outside the flux tubes.

contours C_1 and C_2 , and the flux tube volumes are V_1 and V_2 . The magnetic field is assumed to vanish outside of the flux tubes.

The helicity of the volume V containing the two linked flux tubes is

$$K = \int_V \mathbf{A} \cdot \mathbf{B} d^3r. \quad (11.1)$$

Because $\mathbf{B}$ vanishes outside of the two flux tubes, the helicity integral is finite only inside V_1 and V_2 so Eq. (11.1) reduces to (Moffatt 1978)

$$K = \int_{V_1} \mathbf{A} \cdot \mathbf{B} d^3r + \int_{V_2} \mathbf{A} \cdot \mathbf{B} d^3r. \quad (11.2)$$

The contribution to the helicity from integrating over the volume of flux tube #1 is

$$K_1 = \int_{V_1} \mathbf{A} \cdot \mathbf{B} d^3r. \quad (11.3)$$

In order to evaluate this integral it is recalled that the magnetic flux through a surface S with perimeter C can be expressed as

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} = \oint_C \mathbf{A} \cdot d\mathbf{l}. \quad (11.4)$$

In flux tube #1, $d^3r = d\mathbf{l} \cdot \hat{\mathbf{B}} \Delta S$, where ΔS is the cross-sectional area of flux tube #1 and $d\mathbf{l}$ is an element of length along C_1 . The integrand in Eq. (11.3) can thus be arranged as

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} d^3r &= \mathbf{A} \cdot \mathbf{B} d\mathbf{l} \cdot \hat{\mathbf{B}} \Delta S \\ &= \mathbf{A} \cdot d\mathbf{l} \Phi_1 \end{aligned} \quad (11.5)$$

since $d\mathbf{l}$ is parallel to $\mathbf{B}$ and $B\Delta S = \Phi_1$. Because Φ_1 is by definition constant along the length of flux tube #1, Φ_1 may be factored from the K_1 integral, giving

$$K_1 = \Phi_1 \int_{C_1} \mathbf{A} \cdot d\mathbf{l}. \quad (11.6)$$

However, the flux linked by contour C_1 is precisely the flux in tube #2, i.e., $\int_{C_1} \mathbf{A} \cdot d\mathbf{l} = \Phi_2$ and so

$$K_1 = \Phi_1 \Phi_2. \quad (11.7)$$

Application of the same analysis to flux tube #2 gives $K_2 = \Phi_1 \Phi_2$ and so the total helicity of the two linked flux tubes is

$$K = K_1 + K_2 = 2\Phi_1 \Phi_2. \quad (11.8)$$

The flux tubes did not actually have to be thin: a fat flux tube #2 could be decomposed into a number of adjacent thin flux tubes, in which case K_1 would be Φ_1 times the sum of the flux in all of the thin #2 flux tubes. The helicity is thus just the sum of flux tube linkages since if the two flux tubes each had unit flux, there would be one linkage of flux tube #1 with flux tube #2 and one linkage of flux tube #2 with flux tube #1. If flux tube #2 wrapped around flux tube #1 twice, then the contributions would be $K_1 = 2\Phi_1 \Phi_2$ and $K_2 = 2\Phi_1 \Phi_2$, in which case the helicity would be $K = 4\Phi_1 \Phi_2$. This would correspond to two linkages of flux tube #2 on flux tube #1 and two linkages of flux tube #1 on flux tube #2.

11.2.2 Twist helicity

Now suppose the major radius of flux tube #2 is reduced until flux tube #2 tightly encircles flux tube #1. Then, further suppose, as sketched in Fig. 11.2, that the cross-section of flux tube #2 is both squeezed and stretched until its field lines

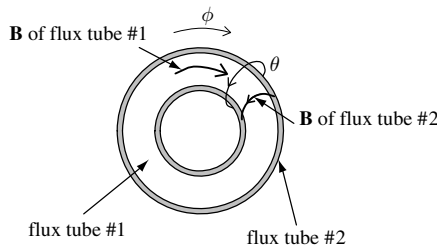


Fig. 11.2 Flux tube #2 deformed so that it tightly encircles flux tube #1 and its cross-section is squeezed and stretched so as to uniformly cover flux tube #1 like a coat of paint (shaded area). Flux in flux tube #1 is Φ , flux in flux tube #2 is $d\psi$.

are uniformly distributed over the length of flux tube #1; the field lines of flux tube #2 are normal to the shaded area in Fig. 11.2. Thus, the volume of flux tube #2 is like a thin coat of paint (shaded region in figure) applied to flux tube #1. Let Φ denote the flux in flux tube #1 and $d\psi$ denote the flux in flux tube #2 so, using Eq. (11.8), the helicity of this configuration is seen to be

$$dK = 2\Phi d\psi. \quad (11.9)$$

We next add another layer of “paint” with more embedded ψ flux, and also with embedded Φ flux so both ψ and Φ increase. The value of Φ can be used to label the layers of “paint” so Φ is the amount of flux in flux tube #1 up to the layer of “paint” labeled by Φ . Furthermore, since ψ increases with added layers of “paint,” ψ must be a function of the layer of “paint” and so $\psi = \psi(\Phi)$. It is therefore possible to write $d\psi = \psi' d\Phi$, where $\psi' = d\psi/d\Phi$. Thus, the amount of helicity added with each layer of “paint” is

$$dK = 2\Phi\psi' d\Phi \quad (11.10)$$

and so the sum of the helicity contributions from all the layers of “paint” is

$$K = 2 \int_0^\Phi \Phi \psi' d\Phi. \quad (11.11)$$

We now show that ψ' represents the twist of the embedded magnetic field. Let ϕ be the angle the long way around flux tube #1 and θ be the angle the long way around flux tube #2 as shown in Fig. 11.2. Thus, increasing ϕ is in the direction of contour C_1 and increasing θ is in the direction of contour C_2 . The perimeter of a cross-section of flux tube #1 is in the θ direction and the perimeter of a cross-section of flux tube #2 is in the ϕ direction. The magnetic field in flux tube #1 can be written as

$$\mathbf{B}_1 = \frac{1}{2\pi} \nabla\Phi \times \nabla\theta = \frac{1}{2\pi} \nabla \times \Phi \nabla\theta, \quad (11.12)$$

which is in the ϕ direction since $\nabla\Phi$ is orthogonal to $\nabla\theta$. Here Φ is the flux linked by a contour going in the direction of $\nabla\theta$. To verify that this is the appropriate expression for $\mathbf{B}_1$, the flux through the cross-section S_1 of flux tube #1 is calculated as follows:

$$\begin{aligned} \text{flux through } S_1 &= \int_{S_1} d\mathbf{s} \cdot \mathbf{B}_1 \\ &= \frac{1}{2\pi} \int_{S_1} d\mathbf{s} \cdot \nabla \times \Phi \nabla\theta \\ &= \frac{1}{2\pi} \int_{C_2} d\mathbf{l} \cdot \Phi \nabla\theta \end{aligned}$$

$$\begin{aligned}
&= \frac{\Phi}{2\pi} \int_{C_2} \mathbf{dl} \cdot \nabla \theta \\
&= \frac{\Phi}{2\pi} \oint d\theta \\
&= \Phi.
\end{aligned} \tag{11.13}$$

The flux Φ can be factored from the integral in the third line above, because Φ is the flux linked by contour C_2 , which goes in the direction of $\nabla\theta$. Similarly, it is possible to write

$$\mathbf{B}_2 = \frac{1}{2\pi} \nabla\psi \times \nabla\phi = \frac{1}{2\pi} \nabla \times \psi \nabla\phi, \tag{11.14}$$

which is in the θ direction since $\nabla\psi$ is orthogonal to $\nabla\phi$. Since $\mathbf{B}_1$ is in the ϕ direction and $\mathbf{B}_2$ is in the θ direction, the total magnetic field can be written as $\mathbf{B} = B_1 \hat{\phi} + B_2 \hat{\theta}$, which is helical.

The twist of the magnetic field is defined as the number of times a field line goes around in the θ direction for one circuit in the ϕ direction. If dl_ϕ is a displacement in the ϕ direction then

$$d\phi = |\nabla\phi| dl_\phi \tag{11.15}$$

and similarly

$$d\theta = |\nabla\theta| dl_\theta. \tag{11.16}$$

The trajectory of a magnetic field line is parallel to the magnetic field so if $\mathbf{dl}$ is an increment along a magnetic field line then $\mathbf{B} \times \mathbf{dl} = 0$ or

$$\frac{dl_\theta}{B_\theta} = \frac{dl_\phi}{B_\phi}. \tag{11.17}$$

Substituting for dl_θ and dl_ϕ using Eqs. (11.15) and (11.16) gives

$$\frac{d\theta}{|\nabla\theta| B_\theta} = \frac{d\phi}{|\nabla\phi| B_\phi}. \tag{11.18}$$

However, $B_\theta = |\nabla\psi| |\nabla\phi| / 2\pi$ and $B_\phi = |\nabla\Phi| |\nabla\theta| / 2\pi$ so

$$\begin{aligned}
\frac{d\theta}{d\phi} &= \frac{|\nabla\theta| B_\theta}{|\nabla\phi| B_\phi} \\
&= \frac{|\nabla\theta| |\nabla\psi| |\nabla\phi| / 2\pi}{|\nabla\phi| |\nabla\Phi| |\nabla\theta| / 2\pi} \\
&= \frac{|\nabla\psi|}{|\nabla\Phi|}.
\end{aligned} \tag{11.19}$$

Finally, if $\psi = \psi(\Phi)$ then $\nabla\psi = \psi'\nabla\Phi$, where $\psi' = d\psi/d\Phi$ and so

$$\frac{d\theta}{d\phi} = \psi'. \quad (11.20)$$

Thus, θ increases ψ' times faster than does ϕ and so if ϕ makes one complete circuit (i.e., goes from 0 to 2π), then θ makes ψ' circuits (i.e., θ goes from 0 to $2\pi\psi'$). The number of times the field line goes around in the θ direction for each time it goes around in the ϕ direction is called the twist

$$T(\Phi) = \psi'. \quad (11.21)$$

Hence Eq. (11.11), which gave the helicity when the ψ flux is embedded in the Φ flux, can be expressed in terms of the twist as

$$K = 2 \int \Phi T(\Phi) d\Phi \quad (11.22)$$

and if the twist is a constant (i.e., $T' = 0$) then

$$K = T\Phi^2. \quad (11.23)$$

11.2.3 Conservation of magnetic helicity during magnetic reconnection

Ideal MHD constrains magnetic flux to be frozen into the frame of the plasma. This means that the topology (connectedness) of magnetic field lines in a perfectly conducting plasma cannot change. As will be shown in Chapter 12, introduction of a small amount of resistivity allows the frozen-in condition to be violated at certain special locations where considerations of geometrical symmetry require the velocity to vanish. At these special locations, the approximation that the $\eta\mathbf{J}$ term in the resistive MHD Ohm's law $\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta\mathbf{J}$ can be neglected so as to obtain the ideal Ohm's law $\mathbf{E} + \mathbf{U} \times \mathbf{B} = 0$ necessarily fails because if $\mathbf{U}$ is zero, the left-hand side has only one term and there is no possibility of balancing this term unless finite resistivity is taken into account. Thus, at these special locations Ohm's law has the form $\mathbf{E} = \eta\mathbf{J}$, in which case magnetic field lines can diffuse across the plasma and reconnect with each other. Although localized to the immediate vicinity of where $\mathbf{U}$ vanishes, magnetic reconnection changes the overall topology of the magnetic field, much like switching railroad tracks at a single crossroads in the middle of a country alters the routing of trains across the entire country. Magnetic reconnection can both destroy and create linkages between flux tubes but, as will be shown, does so in a fashion whereby a replacement linkage is created for every destroyed linkage so that the total number of linkages, and hence the total helicity, is conserved. The ideal MHD constraint of perfect flux conservation is consequently replaced by the somewhat weaker constraint of

having conservation of total magnetic helicity. Reconnection necessarily involves energy dissipation since reconnection requires finite resistivity. Thus, reconnection dissipates magnetic field energy while conserving total magnetic helicity.

Conservation of helicity during reconnection is demonstrated using the sequence shown in Fig. 11.3(a)–(f). An initial state consisting of two linked, untwisted ribbons of magnetic flux is shown in Fig. 11.3(a). If each ribbon is imagined to be a magnetic field line bundle having nominal flux Φ , then according to Eq. (11.8) this initial configuration has helicity $K = 2\Phi^2$. The ribbons are then cut at their line of overlap as in Fig. 11.3(b) and reconnected as in Fig. 11.3(c) to form one long ribbon. This long ribbon in Fig. 11.3(c) is then continuously deformed until in the shape shown in Fig. 11.3(f), a long ribbon having *two* twists. The twist parameter for Fig. 11.3(f) is therefore $T = 2$ and so according to Eq. (11.23) this two-twist ribbon has a helicity $K = T\Phi^2 = 2\Phi^2$. Since the initial and final helicities are both $K = 2\Phi^2$, this demonstrates that reconnection conserves magnetic helicity. Since the magnetic equivalent of cutting ribbons with scissors involves the dissipative process of resistive diffusion of magnetic field across the plasma, it can be concluded that magnetic reconnection conserves helicity, but dissipates magnetic energy.

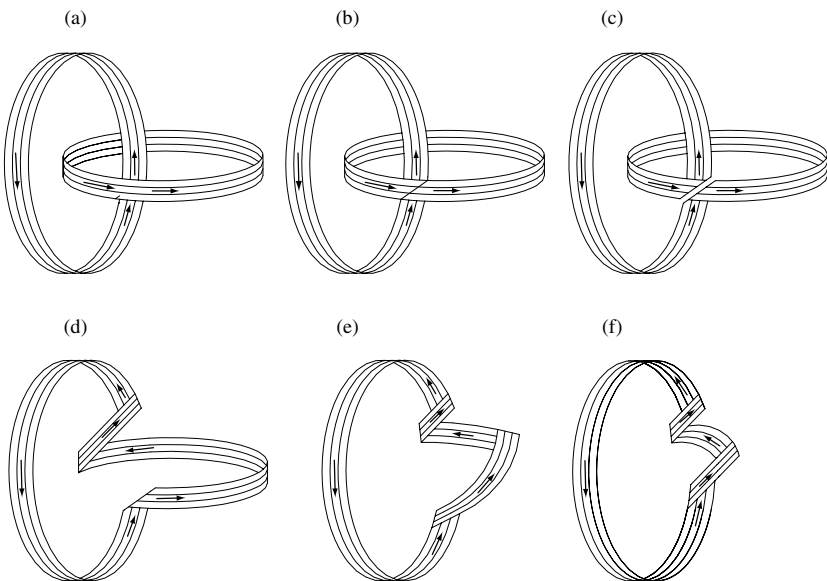


Fig. 11.3 (a) Two linked, untwisted flux tubes squeezed to be ribbons, magnetic field directions shown by arrows; (b) cut at overlay of two ribbons; (c) reconnection at cut; (d) deformation; (e) more deformation; (f) more deformation showing there are two complete twists of magnetic field.

11.3 Woltjer–Taylor relaxation

Woltjer (1958) presented a mathematical proof showing that the lowest energy state of an isolated zero-pressure (i.e., zero β) plasma with a fixed amount of magnetic helicity is a certain kind of force-free state, but did not provide any detailed explanation on how the system might attain this state. Taylor (1974) argued that because reconnection conserves helicity while dissipating magnetic energy, reconnection events ought to provide the operative mechanism whereby an isolated zero β plasma would relax towards a state having the lowest magnetic energy consistent with conservation of helicity. The discussion of Eq. (10.133), the helicity conservation equation, showed that the isolation requirement corresponds to having no field lines penetrate the surface bounding the plasma and also arranging for this surface to be an electrostatic equipotential. In order to investigate the relaxation process, we therefore consider an isolated zero β plasma bounded by a perfectly conducting wall where the normal component of the magnetic field is zero at the wall. Because the wall is a perfect conductor, the tangential electric field and hence the tangential component of the vector potential vanish at the wall. The zero β assumption is an extreme limit of the low- β situation, which corresponds to a plasma where magnetic forces dominate hydrodynamic forces, i.e., the $\mathbf{J} \times \mathbf{B}$ term is much larger than the ∇P term in the MHD equation of motion.

The relaxation process involves minimizing the magnetic energy

$$W = \frac{1}{2\mu_0} \int B^2 d^3r \quad (11.24)$$

subject to the constraint that the total magnetic helicity

$$K = \int \mathbf{A} \cdot \mathbf{B} d^3r \quad (11.25)$$

is conserved (Woltjer 1958, Taylor 1974). The minimum-energy magnetic field for the given helicity is called $\mathbf{B}_{ME}$ and its associated magnetic energy is

$$W_{ME} = \frac{1}{2\mu_0} \int B_{ME}^2 d^3r. \quad (11.26)$$

In order to determine $\mathbf{B}_{ME}$, consider an arbitrary variation $\mathbf{B} = \mathbf{B}_{ME} + \delta\mathbf{B}$ satisfying the same boundary conditions as $\mathbf{B}_{ME}$. By assumption, $\mathbf{B}$ has a higher associated magnetic energy than $\mathbf{B}_{ME}$ since the latter is the minimum energy field. The field $\mathbf{B}$ has an associated vector potential $\mathbf{A} = \mathbf{A}_{ME} + \delta\mathbf{A}$. Because the tangential electric field must vanish at the wall in accordance with the assumption of helicity conservation, the tangential component of $\delta\mathbf{A}$ must vanish at the wall. A naive attempt at minimizing W_{ME} would involve setting $\mathbf{B}_{ME}$ to zero, but such an approach is forbidden because it would make K vanish and violate the helicity

conservation requirement. What must be done then is minimize W subject to the constraint that K remains constant. The variation of the magnetic energy relative to the minimum energy state is

$$\begin{aligned}
 \delta W &= \frac{1}{2\mu_0} \int [(\mathbf{B}_{ME} + \delta \mathbf{B})^2 - B_{ME}^2] d^3r \\
 &= \frac{1}{\mu_0} \int \mathbf{B}_{ME} \cdot \delta \mathbf{B} d^3r \\
 &= \frac{1}{\mu_0} \int \mathbf{B}_{ME} \cdot \nabla \times \delta \mathbf{A} d^3r \\
 &= \frac{1}{\mu_0} \int [\nabla \cdot (\mathbf{B}_{ME} \times \delta \mathbf{A}) + \delta \mathbf{A} \cdot \nabla \times \mathbf{B}_{ME}] d^3r \\
 &= \frac{1}{\mu_0} \int d\mathbf{s} \cdot \mathbf{B}_{ME} \times \delta \mathbf{A} + \int \delta \mathbf{A} \cdot \nabla \times \mathbf{B}_{ME} d^3r \\
 &= \int \delta \mathbf{A} \cdot \nabla \times \mathbf{B}_{ME} d^3r,
 \end{aligned} \tag{11.27}$$

since $d\mathbf{s} \cdot \mathbf{B}_{ME} \times \delta \mathbf{A} = 0$ on the wall. On the other hand, the variation of the total helicity is

$$\begin{aligned}
 \delta K &= \int (\delta \mathbf{A} \cdot \mathbf{B}_{ME} + \mathbf{A}_{ME} \cdot \delta \mathbf{B}) d^3r \\
 &= \int (\delta \mathbf{A} \cdot \mathbf{B}_{ME} + \mathbf{A}_{ME} \cdot \nabla \times \delta \mathbf{A}) d^3r \\
 &= \int (\delta \mathbf{A} \cdot \mathbf{B}_{ME} + \delta \mathbf{A} \cdot \nabla \times \mathbf{A}_{ME}) + \nabla \cdot (\delta \mathbf{A} \times \mathbf{A}_{ME}) d^3r \\
 &= 2 \int \delta \mathbf{A} \cdot \mathbf{B}_{ME} d^3r.
 \end{aligned} \tag{11.28}$$

Minimization of δW subject to the constraint that K remains constant is characterized using a Lagrange multiplier λ so the constrained variational equation is

$$\delta W = \lambda \delta K. \tag{11.29}$$

Substitution for δW and δK gives

$$\int \delta \mathbf{A} \cdot \nabla \times \mathbf{B}_{ME} d^3r = 2\lambda \int \delta \mathbf{A} \cdot \mathbf{B}_{ME} d^3r \tag{11.30}$$

or, after redefining the arbitrary parameter λ ,

$$\int \delta \mathbf{A} \cdot (\nabla \times \mathbf{B}_{ME} - \lambda \mathbf{B}_{ME}) d^3r = 0. \tag{11.31}$$

Since the variation $\delta \mathbf{A}$ is arbitrary within the volume, the quantity in parentheses must vanish and so

$$\nabla \times \mathbf{B}_{ME} = \lambda \mathbf{B}_{ME}. \quad (11.32)$$

The coefficient λ is necessarily constant (i.e., spatially uniform) because the Lagrange multiplier is a constant (see Assignment 2 of Chapter 2, p. 69). Thus, relaxation leads to the same force-free state predicted by Eq. (10.140). These states are a good approximation to many solar and astrophysical plasmas as well as certain laboratory plasmas such as spheromaks and reversed field pinches.

The energy per helicity of a minimum energy state can be written as

$$\begin{aligned} \frac{W}{K} &= \frac{\int B_{ME}^2 d^3r}{2\mu_0 \int \mathbf{A}_{ME} \cdot \mathbf{B}_{ME} d^3r} \\ &= \frac{\int \mathbf{B}_{ME} \cdot \nabla \times \mathbf{A}_{ME} d^3r}{2\mu_0 \int \mathbf{A}_{ME} \cdot \mathbf{B}_{ME} d^3r} \\ &= \frac{\int [\mathbf{A}_{ME} \cdot \nabla \times \mathbf{B}_{ME} + \nabla \cdot (\mathbf{A}_{ME} \times \mathbf{B}_{ME})] d^3r}{2\mu_0 \int \mathbf{A}_{ME} \cdot \mathbf{B}_{ME} d^3r}. \end{aligned} \quad (11.33)$$

However, Eq. (11.32) can be integrated to give

$$\mathbf{B}_{ME} = \lambda \mathbf{A}_{ME} + \nabla f, \quad (11.34)$$

where f is some arbitrary scalar function. This can be used to show

$$\nabla \cdot (\mathbf{A}_{ME} \times \mathbf{B}_{ME}) = \nabla \cdot (\mathbf{A}_{ME} \times \nabla f) = \nabla f \cdot \nabla \times \mathbf{A}_{ME} = \mathbf{B}_{ME} \cdot \nabla f = \nabla \cdot (f \mathbf{B}_{ME}) \quad (11.35)$$

so

$$\int \nabla \cdot (\mathbf{A}_{ME} \times \mathbf{B}_{ME}) d^3r = \int d\mathbf{s} \cdot (f \mathbf{B}_{ME}) = 0, \quad (11.36)$$

since $\mathbf{B}_{ME} \cdot d\mathbf{s} = 0$ by assumption. Thus, Eq. (11.33) reduces to

$$\frac{W}{K} = \frac{\lambda}{2\mu_0} \quad (11.37)$$

indicating that the minimum energy state must have the smallest λ consistent with the prescribed boundary conditions.

11.4 Kinking and magnetic helicity

Magnetic helicity can be manifested in various forms and plasma dynamics can cause transformation from one form to another. We now discuss an important example of helicity-conserving, morphology-altering dynamics, namely the situation where kink instability causes the twist of a straight-axis flux tube to be

transformed into the writhe of the axis of an untwisted flux tube (Berger and Field 1984, Moffatt and Ricca 1992).

Consider an ideal plasma with volume V bounded by a surface S and suppose no magnetic field lines penetrate S so $\mathbf{B} \cdot d\mathbf{s} = 0$ over all of S . The volume V could extend to infinity or be finite; all that is required is $\mathbf{B} \cdot d\mathbf{s} = 0$ on the bounding surface. The helicity contained in V is

$$K_V = \int_V \mathbf{A} \cdot \mathbf{B} d^3r. \quad (11.38)$$

Equation (10.129) showed K_V is gauge-independent because no field lines penetrate S and Eq. (10.134) showed K_V is a conserved quantity, i.e., the total helicity in the volume V is constant.

The volume V is now decomposed into two sub-volumes where $\mathbf{B} \cdot d\mathbf{s} = 0$ everywhere on the surface separating these sub-volumes. Thus, any specific field line in V is entirely in one or the other of the two sub-volumes because if a field line did travel from one sub-volume to the other it would have to penetrate the surface separating these sub-volumes and so violate the assumption $\mathbf{B} \cdot d\mathbf{s} = 0$ on this surface.

- 1 The first sub-volume, called V_{tube} , is a closed flux tube of minor radius a with a possibly helical axis as shown in Fig. 11.4. The axis of this flux tube is concealed in Fig. 11.4, but can be seen in Fig. 11.5 where it is labeled as “helical axis of flux tube.” The length of this flux tube axis is denoted l_{axis} and the variable ξ denotes the distance along this axis from some fixed reference point $\mathbf{x}_0$ on the axis. Incrementing ξ from 0 to l_{axis} thus corresponds to going once around the flux tube axis. A pseudo-angular coordinate ϕ is defined as

$$\phi = 2\pi \frac{\xi}{l_{axis}} \quad (11.39)$$

so that going once around the flux tube axis corresponds to incrementing ϕ from 0 to 2π . The unit vector $\hat{\phi}$ can then be used to define the local direction of the axis. Because the flux tube axis is helical, there exists at each point along the axis a radius of curvature vector $\boldsymbol{\kappa} = \hat{\phi} \cdot \nabla \hat{\phi}$, which is at right angles to $\hat{\phi}$. The local radius of curvature of the axis is $r_{curve} = 1/|\hat{\phi} \cdot \nabla \hat{\phi}|$ (see discussion of Eq. (3.85)). From the point of view of an observer inside the flux tube, the flux tube appears as a long curved tunnel, which eventually closes upon itself. The flux tube minor radius a must always be smaller than r_{curve} ; this restriction corresponds locally to the condition that the major radius of a torus must, by definition, exceed the minor radius. The flux tube interior volume is now imagined to be filled with fiduciary lines aligned parallel to the flux tube axis; the lengths of these fiduciary lines will vary according to their location relative to the axis. We define ξ' as the distance along a fiduciary line of length l' from a plane perpendicular to the flux tube axis at the reference point $\mathbf{x}_0$. The meaning of ϕ can then be extended to indicate the distance along any fiduciary line using the

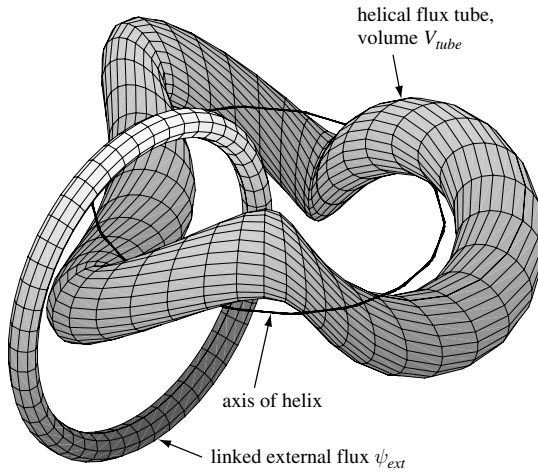


Fig. 11.4 Sub-volume #1 has a volume V_{tube} and is a flux tube with a possibly helical axis. Sub-volume #2 is all space external to sub-volume #1. All field lines in sub-volume #2 linking sub-volume #1 are represented by the thin vertical flux tube labeled ψ_{ext} .

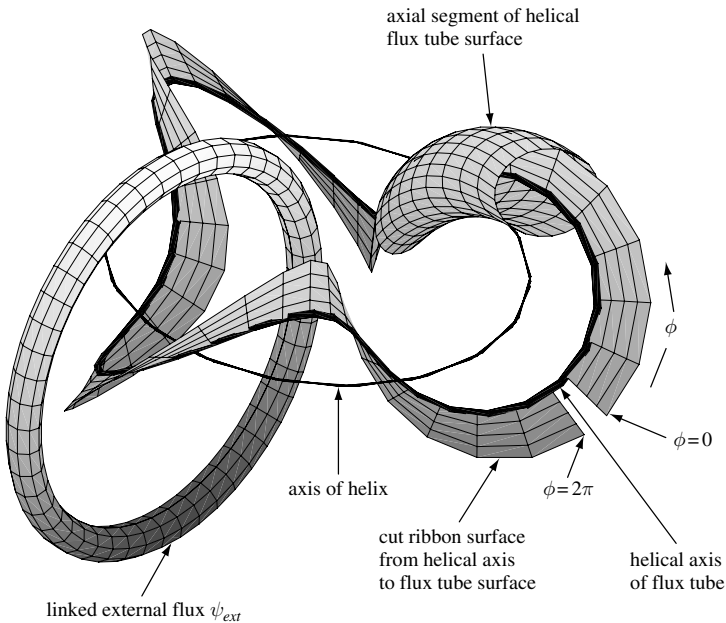


Fig. 11.5 Ribbon surface extending from helical axis of helical flux tube to surface of flux tube with ribbon surface normal to $\hat{\phi} \times \hat{k}$.

relation $\xi' = \phi l' / 2\pi$. Incrementing ϕ from 0 to 2π thus corresponds to going once around any or all of these fiduciary lines and so ϕ provides an unambiguous measure of fractional distance along the flux tube for any point within the flux tube even though the flux tube may be curved, twisted, or helical.

2. The second sub-volume in this decomposition is the remaining volume of V and is labeled as V_{ext} . This second sub-volume may have field lines linking flux tube V_{tube} and the flux due to these linkages is labeled ψ_{ext} . This external flux linkage is represented by the vertical flux tube labeled ψ_{ext} in Fig. 11.4.

If the dynamics within the volume V is governed by ideal MHD then, as mentioned above, the helicity K_V in V must be conserved. Since K_V is a volume integral, the volume of integration can be separated into the flux tube volume and the volume external to the flux tube, i.e., the total helicity can be expressed as the sum

$$K_V = K_{V_{tube}} + K_{V_{ext}}. \quad (11.40)$$

11.4.1 $K_{V_{tube}}$, helicity content of the flux tube

We now consider the helicity content of the possibly helical flux tube,

$$K_{V_{tube}} = \int_{V_{tube}} \mathbf{A} \cdot \mathbf{B} d^3r. \quad (11.41)$$

The magnetic field in the flux tube is decomposed into a component $\mathbf{B}_{axis}$ parallel to the flux tube axis and an orthogonal component $\mathbf{B}_{azimuthal}$, which goes the short way around the axis; thus the magnetic field inside the flux tube is

$$\mathbf{B} = \mathbf{B}_{axis} + \mathbf{B}_{azimuthal}. \quad (11.42)$$

The field lines in the flux tube are assumed to lie in successive layers (magnetic surfaces) wrapped around the flux tube axis. The axial and azimuthal magnetic fields are derived from respective vector potentials $\mathbf{A}_{axis}$ and $\mathbf{A}_{azimuthal}$ so

$$\begin{aligned} \mathbf{B}_{axis} &= \nabla \times \mathbf{A}_{axis} \\ \mathbf{B}_{azimuthal} &= \nabla \times \mathbf{A}_{azimuthal}. \end{aligned} \quad (11.43)$$

These definitions say nothing about the direction of $\mathbf{A}_{axis}$ or $\mathbf{A}_{azimuthal}$ and so, unlike the axisymmetric situation considered when analyzing the Grad–Shafranov equation in Section 9.8.3, here neither $\mathbf{A}_{axis}$ nor $\mathbf{A}_{azimuthal}$ should be construed to be in any particular direction. All that can be said is that the curl of $\mathbf{A}_{axis}$ gives the flux tube axial magnetic field and the curl of $\mathbf{A}_{azimuthal}$ gives the azimuthal field.

The helicity content of the flux tube is thus

$$K_{V_{tube}} = \int_{V_{tube}} (\mathbf{A}_{axis} + \mathbf{A}_{azimuthal}) \cdot (\mathbf{B}_{axis} + \mathbf{B}_{azimuthal}) d^3r \quad (11.44)$$

and this is true even though the axis of the flux tube could be helical.

By definition, each layer of field lines constituting a magnetic surface encloses a flux Φ . An equivalent definition is to state that Φ is the magnetic flux linked by a contour in the magnetic surface going the short way around the flux tube axis. The magnetic surface is labeled by Φ and so Φ can be considered as a coordinate having its gradient always normal to the flux surface. In effect Φ is a rescaled minor radius, since Φ increases monotonically with minor radius.

These definitions are sufficiently general to allow for the axis of the flux tube to be a helix, a knot, or a simple closed curve lying in a plane. If the flux tube axis is just a simple closed curve lying in a plane, then $\nabla \times \hat{\phi}$ is normal to the plane and therefore normal to $\hat{\phi}$. In the slightly more general case where the axis is not in a plane but $\hat{\phi} \cdot \nabla \times \hat{\phi} = 0$, the path traced out by $\hat{\phi}$ can be considered to be the perimeter of some bumpy surface. However, in the most general case where $\hat{\phi} \cdot \nabla \times \hat{\phi}$ is finite, *no surface exists* for which the path traced out by $\hat{\phi}$ is the perimeter (Barnes 1977). There might also be situations where $\hat{\phi} \cdot \nabla \times \hat{\phi} \neq 0$ but the sign of $\hat{\phi} \cdot \nabla \times \hat{\phi}$ alternates. If the average of $\hat{\phi} \cdot \nabla \times \hat{\phi}$ over the length of the axis is zero, then the situation would be similar to the case where $\hat{\phi} \cdot \nabla \times \hat{\phi} = 0$ everywhere, because twists in the axis could be squeezed together axially until mutually canceling out.

Because any magnetic field line in the flux tube lies in some magnetic surface labeled by Φ , any field line in the flux tube has a component parallel to the flux tube axis and possibly also a component perpendicular to the flux tube axis, but never a component in the $\nabla\Phi$ direction. This suggests introduction of an azimuthal coordinate θ , which is defined as the angular distance around the flux tube axis in the $\nabla\phi \times \nabla\Phi$ direction, i.e., θ is defined such that

$$\frac{\nabla\theta}{|\nabla\theta|} = \frac{\nabla\phi \times \nabla\Phi}{|\nabla\phi \times \nabla\Phi|}. \quad (11.45)$$

The local direction of $\nabla\theta$ depends on both ϕ and θ ; furthermore, for θ to denote some definite position, θ must be measured with respect to some unambiguous origin. An unambiguous θ origin can be established by using as a reference the direction of the local radius of curvature vector of the flux tube axis, i.e., the direction of $-\kappa = -\hat{B} \cdot \nabla \hat{B}$ evaluated on the flux tube axis (see discussion of Eq. (9.14) on p. 313). The $\theta = 0$ surface is then a ribbon surface, as shown in Fig. 11.5, that extends from the flux tube axis to the flux tube outer surface and oriented so $\hat{\phi} \times \kappa$ is normal to the ribbon surface. The ribbon is considered to

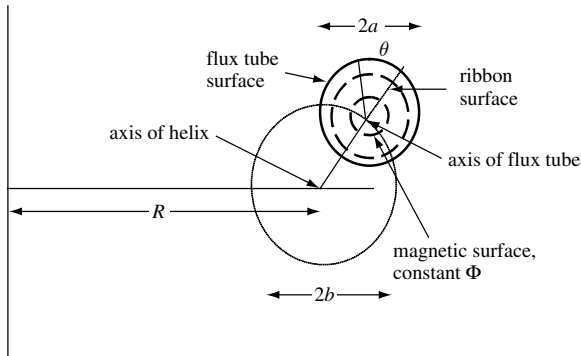


Fig. 11.6 Cross-section showing flux tube of minor radius a , with its axis tracing out a helix with radius b about the “axis of helix.” Interior flux surfaces with $\Phi = \text{const.}$ are shown as dashed lines. The ribbon surface defining $\theta = 0$ is also indicated.

have a cut at $\phi = 0$ in order to distinguish the ribbon ends at $\phi = 0$ and $\phi = 2\pi$ from each other.

Figure 11.6 shows a cross-section of this system. The flux tube has minor radius a and the helical axis of the flux tube traces out a trajectory with minor radius b about an axis that has major radius R . Representative magnetic surfaces (surfaces of constant Φ), the ribbon surface, and the angle θ are also shown in this figure.

The flux Φ was a generalization of the toroidal flux of an axisymmetric system. We now define the corresponding generalization of the poloidal flux $\psi(\Phi)$ to be the magnetic flux penetrating a sub-ribbon extending inwards from the outer surface of the possibly helical flux tube to some given interior magnetic surface Φ . This sub-ribbon is shown in Fig. 11.7 and the definition is such that

$$\begin{aligned}\psi &= \int_{\text{sub-ribbon}} \mathbf{ds} \cdot \mathbf{B} \\ &= \int_{\text{sub-ribbon}} \mathbf{ds} \cdot \nabla \times \mathbf{A} \\ &= \oint_C \mathbf{dl} \cdot \mathbf{A},\end{aligned}\tag{11.46}$$

where the contour C follows the perimeter of the sub-ribbon and specifically follows the two ends at $\phi = 0$ and $\phi = 2\pi$. This definition implies $\psi = 0$ on the outer surface of the flux tube because the sub-ribbon area is zero at this location.

We now define the vector potential

$$\mathbf{A}_{\text{azimuthal}} = \frac{\psi(\Phi)}{2\pi} \nabla \phi\tag{11.47}$$

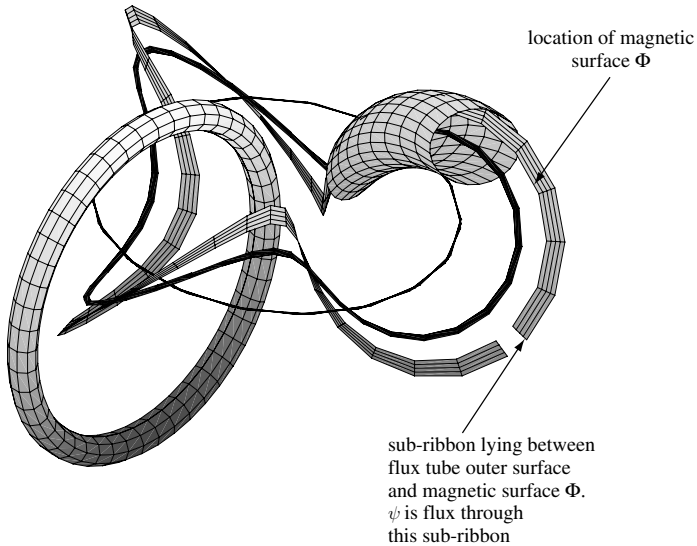


Fig. 11.7 ψ is flux through cut sub-ribbon located between magnetic surface Φ and outer surface of flux tube.

and note that this definition is valid only in the range $0 \leq \phi < 2\pi$. The validity of this definition is established by calculating the line integral of the presumed vector potential following the contour C around the perimeter of the sub-ribbon to obtain

$$\begin{aligned}
 \oint_C \mathbf{dl} \cdot \mathbf{A}_{azimuthal} &= \oint_C \mathbf{dl} \cdot \frac{\psi(\Phi)}{2\pi} \nabla \phi \\
 &= \frac{\psi(\Phi)}{2\pi} \int_{C_\Phi} \mathbf{dl} \cdot \nabla \phi \\
 &= \psi(\Phi),
 \end{aligned} \tag{11.48}$$

where $\mathbf{dl} \cdot \nabla \phi = d\phi$. The justification for the steps in Eq. (11.48) is as follows: Figure 11.7 shows that the contour C around the perimeter of the sub-ribbon consists of four segments, namely a segment following the outside edge of the sub-ribbon (this segment lies in the outer surface of the flux tube where $\psi = 0$), the segment labeled C_Φ on the inside edge of the sub-ribbon, a segment at the $\phi = 0$ end, and an oppositely directed segment at the $\phi = 2\pi_-$ end. Only the C_Φ segment contributes to the integral because (i) $\mathbf{dl} \cdot \nabla \phi = 0$ on the ends $\phi = 0$ and $\phi = 2\pi_-$ since ϕ is constant on both these ends and (ii) $\psi = 0$ on the outside edge of the sub-ribbon.

The magnetic field inside the flux tube can thus be written as

$$\mathbf{B} = \mathbf{B}_{axis} + \nabla \times \frac{\psi(\Phi)}{2\pi} \nabla \phi, \quad (11.49)$$

where $\mathbf{B}_{axis}$ is the magnetic field component parallel to the flux tube axis. This decomposition of the magnetic field is valid even for a twisted or knotted axis provided it is used only for situations where $0 \leq \phi < 2\pi$.

The vector potential for the entire field can be written as

$$\mathbf{A} = \mathbf{A}_{axis} + \frac{\psi(\Phi)}{2\pi} \nabla \phi. \quad (11.50)$$

These results can be used in Eq. (11.44) to give the helicity content of the flux tube as

$$\begin{aligned} K_{V_{tube}} &= \int_{V_{tube}} d^3r \left(\mathbf{A}_{axis} + \frac{\psi(\Phi)}{2\pi} \nabla \phi \right) \cdot \left(\mathbf{B}_{axis} + \frac{1}{2\pi} \nabla \psi \times \nabla \phi \right) \\ &= \int_{V_{tube}} d^3r \left(\mathbf{A}_{axis} \cdot \mathbf{B}_{axis} + \frac{\psi(\Phi)}{2\pi} \nabla \phi \cdot \mathbf{B}_{axis} + \mathbf{A}_{axis} \cdot \frac{1}{2\pi} \nabla \psi \times \nabla \phi \right). \end{aligned} \quad (11.51)$$

This expression may now be decomposed into writhe and twist terms,

$$K_{V_{tube}} = K_{writhe} + K_{twist}$$

respectively defined as

$$K_{writhe} = \int_{V_{tube}} d^3r \mathbf{A}_{axis} \cdot \mathbf{B}_{axis} \quad (11.52)$$

and

$$K_{twist} = \int_{V_{tube}} d^3r \left(\frac{\psi(\Phi)}{2\pi} \nabla \phi \cdot \mathbf{B}_{axis} + \mathbf{A}_{axis} \cdot \frac{1}{2\pi} \nabla \psi \times \nabla \phi \right). \quad (11.53)$$

As will be shown below, K_{twist} depends on ψ being finite whereas K_{writhe} depends on the extent to which the flux tube axis is helical. The flux tube helicity can thus be entirely due to K_{writhe} , entirely due to K_{twist} , or due to some combination of these two types of helicity.

11.4.2 Evaluation of K_{twist}

Since the K_{twist} integral is a *volume* integral, the flux tube may be cut at $\phi = 0$ without affecting this integral. Making such a cut means that ϕ is restricted to the range $0 \leq \phi < 2\pi$ and therefore does not make a complete circuit around the axis of the flux tube. The evaluation of this integral is insensitive to the connectivity

of the axis because connectivity is a concept that makes sense only after making a complete circuit of the axis. The axial magnetic field within the cut volume may be expressed as

$$\mathbf{B}_{axis} = \frac{1}{2\pi} \nabla \times \Phi \nabla \theta, \quad (11.54)$$

where θ is the angular distance on a contour C_θ encircling the flux tube axis and lies in a magnetic surface. This representation for the axial magnetic field is appropriate here since at each ϕ for $0 \leq \phi < 2\pi$ we may write

$$\begin{aligned} \Phi &= \int \mathbf{ds} \cdot \mathbf{B}_{axis} \\ &= \frac{1}{2\pi} \int \mathbf{ds} \cdot \nabla \times \Phi \nabla \theta \\ &= \frac{1}{2\pi} \oint_{C_\theta} \mathbf{dl} \cdot \Phi \nabla \theta \\ &= \frac{\Phi}{2\pi} \oint_{C_\theta} \mathbf{dl} \cdot \nabla \theta \\ &= \frac{\Phi}{2\pi} \oint_{C_\theta} d\theta. \end{aligned} \quad (11.55)$$

By uncurling Eq. (11.54) it is seen that vector potential associated with the axial magnetic field may be represented in the cut flux tube as

$$\mathbf{A}_{axis} = \frac{\Phi}{2\pi} \nabla \theta, \quad 0 \leq \phi < 2\pi \quad (11.56)$$

so

$$\begin{aligned} K_{twist} &= \frac{1}{4\pi^2} \int_{V_{tube}} d^3r (\psi \nabla \phi \cdot \nabla \Phi \times \nabla \theta + \Phi \nabla \theta \cdot \nabla \psi \times \nabla \phi) \\ &= \frac{1}{4\pi^2} \int_{V_{tube}} d^3r (\psi \nabla \phi \cdot \nabla \Phi \times \nabla \theta + \Phi \psi' \nabla \theta \cdot \nabla \Phi \times \nabla \phi) \\ &= \frac{1}{4\pi^2} \int_{V_{tube}} d^3r (-\psi + \Phi \psi') \nabla \theta \cdot \nabla \Phi \times \nabla \phi. \end{aligned} \quad (11.57)$$

The three direction gradients $\nabla \theta$, $\nabla \Phi$, and $\nabla \phi$ form an orthonormal coordinate system and an element of volume in this system is given by

$$d^3r = dl_\phi dl_\theta dl_\phi = \frac{d\Phi d\theta d\phi}{\nabla \theta \cdot \nabla \Phi \times \nabla \phi} \quad (11.58)$$

since $d\theta = dl_\theta |\nabla\theta|$, $d\phi = dl_\phi |\nabla\phi|$, and $d\Phi = dl_\Phi |\nabla\Phi|$. Thus,

$$\begin{aligned}
 K_{twist} &= \frac{1}{4\pi^2} \int_0^\Phi d\Phi \int_{\theta=0}^{2\pi} d\theta \int_{\phi=0}^{(2\pi)-} d\phi (-\psi + \Phi\psi') \\
 &= - \int_0^\Phi \psi d\Phi + \int_0^\Phi \psi' \Phi d\Phi \\
 &= - \int_0^\Phi \psi d\Phi + \int_0^\psi \Phi d\psi \\
 &= 2 \int_0^\Phi \Phi \frac{d\psi}{d\Phi} d\Phi
 \end{aligned} \tag{11.59}$$

where the integration limit $(2\pi)_-$ corresponds to being infinitesimally less than 2π . The last line has been obtained using the relationship $\int d(\psi\Phi) = [\psi\Phi]_{axis}^{surface} = \int \Phi d\psi + \int \psi d\Phi$ and noting that the integrated term vanishes since $\psi = 0$ on the flux tube surface and $\Phi = 0$ on the flux tube axis. Since Eq. (11.59) is consistent with the definition of twist discussed in reference to Eqs. (11.21) and (11.22), the decision to use the name K_{twist} for the helicity term defined in Eq. (11.53) is validated.

11.4.3 Evaluation of K_{writhe}

Connectivity of the flux tube is the important issue here. In order to evaluate K_{writhe} the volume element is now expressed as

$$d^3r = d\mathbf{l} \cdot d\mathbf{s}, \tag{11.60}$$

where $d\mathbf{l}$ is an increment of length along the axis and $d\mathbf{s}$ is an element of surface in the plane perpendicular to the axis so $\mathbf{B}_{axis} \cdot d\mathbf{s} = d\Phi$. Because the line integral involves a *complete circuit* of the flux tube axis, in contrast to the earlier evaluation of K_{twist} , we now avoid using the gradient of a scalar to denote distance along the axis. Using Eq. (11.60) in Eq. (11.52) the writhe helicity may be expressed as

$$\begin{aligned}
 K_{writhe} &= \int_{C_{axis}} \mathbf{A}_{axis} \cdot d\mathbf{l}_{axis} \int d\Phi \\
 &= \Phi \int_{C_{axis}} \mathbf{A}_{axis} \cdot d\mathbf{l}_{axis}.
 \end{aligned} \tag{11.61}$$

This integral differs topologically from the integrals of the previous section, because here the contour is a *complete circuit*, i.e., ϕ varies from 0 to 2π and not from 0 to $2\pi_-$.

A contour path C of an integral $\int_C \mathbf{A} \cdot d\mathbf{l}$ may be continuously deformed into a new contour path C' without changing the value of the integral, provided no magnetic flux is linked by the surface S bounded by C and C' . This property is

a three-dimensional analog to the concept of analyticity for a contour integral in the complex plane and its validity is established as follows: since no magnetic flux is assumed to link the surface S bounded by C and C' we may write

$$\begin{aligned} 0 &= \int_S \mathbf{B} \cdot d\mathbf{s} \\ &= \oint \mathbf{A} \cdot d\mathbf{l} \\ &= \int_C \mathbf{A} \cdot d\mathbf{l} - \int_{C'} \mathbf{A} \cdot d\mathbf{l}. \end{aligned} \quad (11.62)$$

The presumption that it is possible to continuously deform C into C' imposes the requirement that a surface S exists between C and C' ; this is only true if C does not link C' .

We now consider some limiting cases for K_{writhe} and to aid in visualization it should be recalled that the flux tube axis is assumed to have a helical trajectory with helix major radius R and helix minor radius b as shown in Fig. 11.6.

Limiting case where flux tube axis is not helical

This situation corresponds to having $b = 0$, in which case the flux tube axis lies in a plane. The contour C_{axis} in Eq. (11.61) can thus be slipped from its original position through the $\mathbf{B}_{axis}$ field lines to the surface of the flux tube without crossing any $\mathbf{B}_{axis}$ field lines. This is topologically possible because, by definition, all $\mathbf{B}_{axis}$ field lines in the flux tube are parallel to the flux tube axis. Thus, here

$$K_{writhe} = \Phi \int_{C'} \mathbf{A}_{axis} \cdot d\mathbf{l}_{axis}, \quad (11.63)$$

where C' lies on the flux tube surface. The line integral C' encircles the external flux linked by the flux tube and so

$$K_{writhe} = \Phi \psi_{ext}, \quad (11.64)$$

when the flux tube axis is not helical.

When the flux tube axis is helical, two possibilities exist, namely $a < b$ and $a > b$. The distinction between the $a < b$ and $a > b$ cases is sketched in Fig. 11.8. The parameter b characterizes the amplitude of the kink so infinitesimal b corresponds to a very weak kinking of the flux tube.

Strong kink case: axis is helical and $b > a$

Because the helix minor radius b exceeds the flux tube minor radius a here, the entire flux tube revolves around the axis of the helix as shown in Fig. 11.9. Because the helical axis closes upon itself, it must have an integral number of periods and we let N be the number of helix periods. However, because $\mathbf{B}_{axis}$ is everywhere parallel to the flux tube axis, the flux tube axis may again be slipped

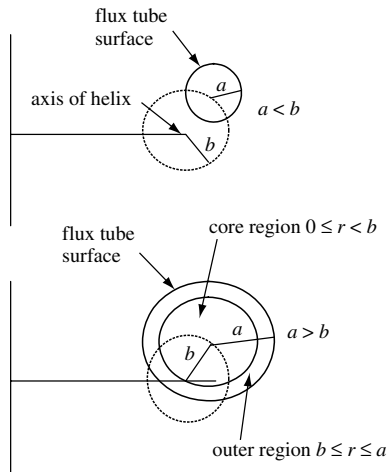


Fig. 11.8 Top: case where flux tube radius a is less than radius b of helix traced out by flux tube axis. Bottom: case where $a > b$; the core region $0 \leq r < b$ rotates around the axis of the helix whereas the outer region with $b < r < a$ wobbles about the axis of the helix.

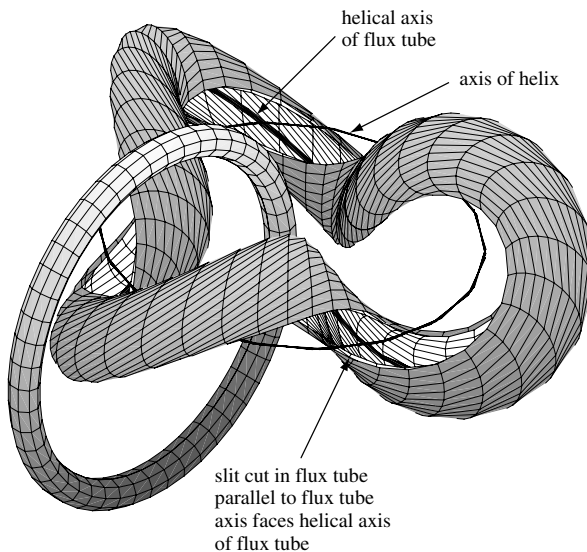


Fig. 11.9 Helical axis of flux tube may be slipped through $\mathbf{B}_{axis}$ field lines of flux tube to coincide with axis of helix (note that open slit in flux tube always faces axis of helix).

through the $\mathbf{B}_{axis}$ field lines and in this case moved towards the helix axis until it coincides with the helix axis C_{helix} . The writhe helicity can thus be expressed as

$$K_{writhe} = \Phi \int_{C_{helix}} \mathbf{A}_{axis} \cdot d\mathbf{l}_{axis}. \quad (11.65)$$

However, the flux tube links the helix axis N times and so

$$\int_{C_{helix}} \mathbf{A}_{axis} \cdot d\mathbf{l}_{axis} = N\Phi + \psi_{ext}. \quad (11.66)$$

Thus,

$$K_{writhe} = N\Phi^2 + \Phi\psi_{ext} \quad \text{if } b > a. \quad (11.67)$$

Hence, the helicity of the flux tube is

$$\begin{aligned} K_{V_{tube}} &= K_{twist} + K_{writhe} \\ &= 2 \int \Phi \frac{d\psi}{d\Phi} d\Phi + N\Phi^2 + \Phi\psi_{ext} \quad \text{if } b > a. \end{aligned} \quad (11.68)$$

Weak kink case: flux tube axis is helical but $a > b$

In this situation the flux tube may be subdivided into an inner core with minor radius $r < b$ and associated flux $\Phi b^2/a^2$ and an outer annular region with $b < r < a$ with the remainder of Φ (to keep matters simple, we are assuming that Φ is evenly distributed over the minor cross-section of the flux tube and so the flux in the inner core is proportional to the normalized cross-sectional area of the inner core). We again slip the flux tube axis through the parallel $\mathbf{B}_{axis}$ field lines to be coincident with the helix axis. However, now only the inner core of the flux tube ($r < b_{axis}$) rotates around the helix axis. The outer annular region of the flux tube merely wobbles around the helix axis and does not link the helix axis. Thus, the linked flux is

$$\int_{C_{helix}} \mathbf{A}_{axis} \cdot d\mathbf{l}_{axis} = N\Phi b^2/a^2 + \psi_{ext} \quad (11.69)$$

and so the writhe helicity is

$$K_{writhe} = N\Phi^2 b^2/a^2 + \Phi\psi_{ext} \quad \text{if } a > b. \quad (11.70)$$

Hence, the helicity of the flux tube is

$$\begin{aligned} K_{V_{tube}} &= K_{twist} + K_{writhe} \\ &= 2 \int \Phi \frac{d\psi}{d\Phi} d\Phi + N\Phi^2 b^2/a^2 + \Phi\psi_{ext} \quad \text{if } a > b. \end{aligned} \quad (11.71)$$

11.4.4 Twist and writhe exchange in a kink instability

A kink instability is governed by ideal MHD and so must be helicity-conserving. The number of turns of the kink is defined by the initial twist condition $\mathbf{k} \cdot \mathbf{B} = k_\phi B_\phi + k_\theta B_\theta = 0$ so if $k_\theta = 1/a$ and $k_\phi = n/R$ then the kink will result in a helix with $N = n$. Since helicity is conserved, it is seen that as the amplitude of the kink increases (i.e., b increases), the flux tube twist $d\psi/d\Phi$ will have to decrease. In particular if the twist is uniform so that $\psi = T\Phi$, where twist T is independent of Φ , then as in Section 11.2.2 we may write

$$2 \int \Phi \frac{d\psi}{d\Phi} d\Phi = T\Phi^2 \quad (11.72)$$

and so from conservation of helicity we conclude

$$T + Nb^2/a^2 = \text{const. for } b < a; \quad (11.73)$$

this is an example of the Calugareanu theorem (Calugareanu 1959).

A kink instability that starts with $b = 0$ and an initial twist $T_{\text{initial}} = n$ will grow until at $b = a$ the twist vanishes and all helicity is contained by an N -turn writhe where $N = T_{\text{initial}}$. Thus, a kink will convert twist into writhe so in the strong kink case (i.e., $b > a$) the result is $T = 0$ and $N = T_{\text{initial}}$. This property can be confirmed by taking a length of garden hose with a stripe running along the length and connecting the two ends together to form a closed flux tube with a stripe running along the axis. Initially, $N + T = 0$ because the stripe is not twisted and the hose axis is not helical. If the hose is deformed into a right-handed helix, then the stripe will make a left-handed helix about the hose axis to keep $N + T = 0$. If the stripe is initially a right-handed helix when the hose is not a helix, then deformation of the hose into a right-handed helix will result in the stripe becoming parallel to the hose axis.

11.5 Assignments

1. Derive the helicity conservation equation for a resistive plasma

$$\frac{dK}{dt} + \int_S d\mathbf{s} \cdot (\varphi \mathbf{B} + \mathbf{E} \times \mathbf{A}) = -2 \int d^3r \, \eta \mathbf{J} \cdot \mathbf{B} \quad (11.74)$$

and express it in the form

$$\frac{dK}{dt} + \int_S d\mathbf{s} \cdot \left(2\varphi \mathbf{B} + \mathbf{A} \times \frac{\partial \mathbf{A}}{\partial t} \right) = -2 \int d^3r \, \eta \mathbf{J} \cdot \mathbf{B}, \quad (11.75)$$

where φ is the electrostatic potential. What happens to the surface integral when the surface is a perfectly conducting wall? Hint: start by evaluating $-\partial(\mathbf{A} \cdot \mathbf{B})/\partial t$ and make repeated use of Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$. The electric field should be expressed as $\mathbf{E} = -\nabla \varphi - \partial \mathbf{B}/\partial t$ and use the resistive MHD Ohm's law $\mathbf{E} + \mathbf{U} \times \mathbf{B} = \eta \mathbf{J}$.

2. Alternative explanation for why helicity is conserved better than magnetic energy:

- (a) By subtracting $\mathbf{E}$ dotted with Ampère's law from $\mathbf{B}$ dotted with Faraday's law derive the MHD limit of Poynting's theorem:

$$\frac{\partial}{\partial t} \left(\frac{B^2}{2\mu_0} \right) + \nabla \cdot \left(\frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \right) = -\mathbf{E} \cdot \mathbf{J}. \quad (11.76)$$

- (b) Consider a plasma in a perfectly conducting chamber with no vacuum gap between the plasma and the chamber wall. Show that integration of Eq. (11.74) and Eq. (11.76) over the entire volume respectively give

$$\frac{dK}{dt} = -2 \int d^3r \eta \mathbf{J} \cdot \mathbf{B} \quad (11.77)$$

and

$$\frac{dW}{dt} = - \int d^3r \eta J^2, \quad (11.78)$$

where $W = \int d^3r B^2/2\mu_0$ is the magnetic energy.

- (c) Show that the right-hand side of Eq. (11.78) is proportional to higher order spatial derivatives of $\mathbf{B}$ than is the right-hand side of Eq. (11.77). Use this property to argue that the rate of dissipation of magnetic energy greatly exceeds that of magnetic helicity when the dynamics is spatially complex so that most of the spectral power is in short characteristic scale lengths.
- (d) Explain why this difference in dissipation rates could be approximated by assuming that magnetic helicity remains constant during magnetic energy decay via dynamics having fine scales (e.g., localized reconnection). Argue that the decay of magnetic energy is thereby constrained by the requirement that helicity is conserved.
3. Show that the minimum-energy state given by Eq. (11.32) is a force-free configuration. Is λ spatially uniform? Is this the same result as given in Eqs. (10.139) and (10.140)? Take the curl again to obtain

$$\nabla^2 \mathbf{B} + \lambda^2 \mathbf{B} = 0.$$

What are the components of this equation in axisymmetric cylindrical geometry? Be careful when evaluating the components of $\nabla^2 \mathbf{B}$ to take into account derivatives operating on unit vectors (e.g., $\nabla^2 (B_\phi \hat{\phi}) \neq \hat{\phi} \nabla^2 B_\phi$, see Appendix B).

- (a) Show that for axisymmetric cylindrical geometry the minimum-energy states have the magnetic field components

$$B_z(r) = \bar{B} J_0(\lambda r)$$

$$B_\phi(r) = \bar{B} J_1(\lambda r).$$

Sketch $B_z(r)$ and $B_\phi(r)$. This is called the Bessel function model or Lundquist solution (Lundquist 1950) to the force-free equation and is often an excellent representation for nearly force-free equilibria such as spheromaks, reversed field pinches, and solar coronal loops.

4. For a cylindrical system with coordinates $\{r, \phi, z\}$ show that if χ satisfies the Helmholtz equation

$$\nabla^2 \chi + \lambda^2 \chi = 0$$

then

$$\mathbf{B} = \lambda \nabla \chi \times \nabla z + \nabla \times (\nabla \chi \times \nabla z)$$

is a solution of the force-free equation $\mu_0 \mathbf{J} = \lambda \mathbf{B}$. By assuming χ is independent of ϕ and is of the form $\chi \sim f(r) \cos kz$, find $f(r)$ and then determine the components of $\mathbf{B}$. Show that if λ satisfies an eigenvalue condition, it is possible to have a finite force-free field having no normal component on the walls of a cylinder of length h and radius a . Give the magnetic field components for this situation (spheromak model). Calculate the poloidal flux $\psi(r, z)$ by direct integration of $B_z(r, z)$ and plot surfaces of constant poloidal flux. Hint: the answers will be in terms of Bessel functions J_0 and J_1 .

5. Obtain a ribbon such as is used in gift-wrapping, make one complete twist in this ribbon and tape the ends together. By manipulating the twisted ribbon (Pfister and Gekelman 1991, Bellan 2000) show the following:
- If the ribbon is manipulated to have no twists, then it has a figure-eight pattern with a cross-over. If the flux through the ribbon is Φ , show that the helicity of one full twist is Φ^2 . Then argue that the helicity of one cross-over must also be Φ^2 and by drawing arrows on the ribbons show that there are two distinct types of cross-over corresponding to left- and right-handed helicity respectively.
 - Cut the ribbon all the way along its length so it becomes two interlinked ribbons of unequal width. Assuming the original ribbon had a flux Φ , calculate the helicity of the new configuration taking into account the helicity due to both twists and linkages and show that helicity is conserved.
6. Demonstrate that the sum of twist plus writhe is conserved using two ropes with radius of approximately 1 cm having distinct colors and a broomstick with a radius of approximately 2 cm. Wrap one rope around the other four times in a left-handed sense to form a simple braid of the two ropes. The wrapping should have a long pitch so that the distance between turns is 10–20 cm. Then take this braid and wrap it in a left-handed sense around the broomstick. What happens to the wrapping of the one rope around the other? Explain this in terms of twist and writhe helicity and conservation of helicity. Next start with the two ropes parallel to each other and wrap them four times in a left-handed sense around the broomstick. Check to see if the ropes become wrapped around each other and, if so, in what sense are they wrapped around each other and how many times? Explain your results in terms of twist, writhe, and cross-over helicity.